



Okan Berhoğlu

Robotics Engineer

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Profile

Robotics Engineer with hands-on experience in robotics research, industrial software development, and autonomous systems. Highly skilled in robot software, industrial communication, data acquisition, simulation, and system integration, with proven experience building and deploying automation and digitalization solutions. Mechanical Engineering background with an M.S. in Robotics.

Education

Middle East Technical University

Ankara, Turkey
Master of Science, Robotics
2024 – 2026

TOBB Economics and Technology University

Ankara, Turkey
Bachelor of Science, Mechanical Engineering
2018 – 2023

Atatürk University

Remote
Associate Degree as Second Degree, Computer Programming
2021 – 2023

Skills

Programming Languages: Java, C#, Python, C++

Robotics & Simulation: ROS/ROS2, MATLAB, Simulink, Simscape, Unity, System Modelling and Identification

Industrial & Data Systems: Industrial Machine Communication, SQL, InfluxDB

Tools & Platforms: Git, Linux, CUDA, SolidWorks

Languages: Turkish – Native, English – Upper-Intermediate: TOEFL iBT 2026 score: 5/6 (equivalent to 100/120 in old score system)

Work Experience

Middle East Technical University Ankara, Turkey

Researcher October 2025 – Present

- Conduct research on mobile robotic systems, including kinematic and dynamic modeling, and autonomous navigation algorithms development.
- Contribute to a research project investigating human rhythmic behavior. Within the project, I developed an experimental setup using a 3-axis haptic device, collected data from predefined rhythmic motions, and conducted system identification and behavior modeling studies.

Teknopar Industrial Automation Ankara, Turkey

Software Engineer February 2024 – October 2025, 1 year 9 months

- Built a Windows service-based data acquisition application in C# for a wide range of factory machines, including CNC controllers, robots, and other industrial equipment, each with distinct communication protocols and integration complexity, by implementing TCP/UDP-based industrial communication and standardizing data into an MTConnect-like format, enabling real-time factory floor visibility for Teknopar's TIA Platform.
- Developed an OSGi application using the Java programming language, with a Swing user interface, for military truck systems, enabling operators to monitor and control onboard power units and the vehicle's electrical system in real time, by integrating industrial communication protocols including EtherNet/IP and Modbus TCP for device communication and data collection.
- Secured and delivered a TÜBİTAK funded project by co-authoring the technical proposal covering system modeling, autonomy stack, and digital twin design for a mobile robot and led the technical implementation of AI-based environmental perception and autonomous navigation using ROS2, PyTorch, 3D LiDAR, and depth camera sensors.

Candidate Engineer December 2022 – February 2024, 1 year 3 months

- Supported R&D efforts by collecting and analyzing data from CNC machines and industrial systems to inform data acquisition design.
- Contributed to modeling an electromechanical system in Simulink and Simscape for a defense industry project, as part of a team integrating it with IBM Rhapsody so systems engineers could assess design changes against requirements without manual recalculation. Also contributed to validating the model against real-world hardware data through grey-box system identification to make the model more realistic.

Intern May 2022 – December 2022, 8 months

- Assisted early-stage industrial communication studies by researching data acquisition methods for CNC machines and industrial equipment.
- Simulated and analyzed real-world motion of an electromechanical system for a defense industry project by developing a MATLAB Simscape Multibody model that integrated CAD geometry with dynamic parameters.
- Created 3D models of hydraulic system components in SolidWorks, integrating them into the full system assembly to support customer presentations and downstream design of related parts.

Anova R&D Technologies Ankara, Turkey

Intern May 2021 – August 2021, 4 months

- Gained hands-on experience in machining, quality control, and finishing processes, and contributed to mold and fixture design activities.